

There’s Nothing in the Air

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Abstract

Why do wages grow faster in bigger cities? We use French administrative data to decompose the urban wage growth premium and find that the answer has surprisingly little to do with cities themselves. While we document substantially faster wage growth in larger cities, 80% of the premium disappears after controlling for the composition of firms and coworkers. We also document significantly higher job-to-job transition rates in larger cities, suggesting workers climb the job ladder faster. Most strikingly, when we focus on workers who remain in the same job – eliminating the job ladder mechanism – the urban wage growth premium falls by 94.1% after accounting for firms and coworkers. The residual effect is statistically indistinguishable from zero. These results challenge the view that cities generate human capital spillovers “in the air,” suggesting instead that urban wage dynamics reflect the sorting of firms and workers and the pace of job mobility.

JEL codes: R3, J2

Keywords: Agglomeration, Sorting, City size, Learning

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Introduction

We have known, at least since [Smith \(1776\)](#), that the returns to labor differ across the city size distribution, and that these differences are deeply intertwined with the sorting of workers across space:

Industry, therefore, must be better rewarded, the wages of labour and the profits of stock must evidently be greater, in the one situation than in the other [town vs. country]. But stock and labour naturally seek the most advantageous employment. They naturally, therefore, resort as much as they can to the town, and desert the country.

One of the central tasks of urban and regional economics has been to understand why wages are higher in bigger cities. Does it reflect spillovers that become stronger in dense cities, or does it reflect sorting of firms and workers? The answer to this question has important implications for the design of place-based policies and land-use regulations.

Famously, [Marshall \(1890\)](#) argued that “*the mysteries of the trade become no mysteries; but are as it were in the air*”, implying ambient learning spillovers from neighbors. This has become somewhat of a canonical explanation for why wages are higher in bigger cities. In this paper, we revisit the urban wage growth premium and provide new evidence that it is almost entirely mediated by the firms and coworkers that sort into cities.

In particular, we are motivated by a burgeoning literature on the role of coworkers in within-firm human capital accumulation ([Nix, 2020](#), [Jarosch, Oberfield, and Rossi-Hansberg, 2021](#), [Gregory, 2023](#), [Herkenhoff, Lise, Menzio, and Phillips, 2024](#), [Adenbaum, Babalievsky, and Jungerman, 2024](#)) which has found that a substantial fraction of on-the-job learning is attributed to having more-skilled coworkers. A natural implication of these findings is that, since bigger cities tend to have more productive firms and more skilled workers, sorting of firms and coworkers into bigger cities may be able to explain the dynamic urban wage premium documented in the seminal works of [Glaeser and Maré \(2001\)](#) and [De la Roca and Puga \(2017\)](#).

We focus on France and use rich matched employer-employee administrative data that allow us to track workers’ labor markets trajectories over time to first document that the urban wage premium is significant: doubling the population of a commuting zone is associated with a 0.17€ increase in hourly wage growth. We then systematically decompose this premium into several components. First, we show that when we control for firm fixed effects, the dynamic urban wage premium falls by 61.5%, suggesting that sorting of firms into cities

accounts for a large share of the dynamic urban wage premium. Second, we show that adding detailed controls for coworkers, mirroring the functional forms in [Jarosch, Oberfield, and Rossi-Hansberg \(2021\)](#) and [Adenbaum, Babalievsky, and Jungerman \(2024\)](#), without the firm fixed effects, similarly reduces the urban wage premium by 73.1%. Adding both these coworker terms and the firm fixed effects explains 79.3%.

Next, we provide evidence that what remains of the premium is driven by job mobility and labor market dynamics rather than human capital growth. We show that job transitions occur more frequently in big cities: doubling city population is associated with a 2% increase in the rate of job transitions. We then redo our decomposition restricting the sample to workers that did not switch jobs, allowing us to control for these differences across cities. We find that controlling for firm fixed effects and coworkers now explains 94.1% of the urban wage premium.

This is our main finding: adequately controlling for differences in firms, coworkers, and job mobility across cities almost entirely explains the urban wage growth premium. The residual effect after controlling for these three channels is both statistically and economically insignificant. Insofar as wages grow faster in bigger cities, our results suggest it arises from differences in distributions of firms and coworkers, as well as heterogeneity in job mobility patterns. These results shed new light on the underlying mechanisms behind the measured urban wage growth premium and leave little room for ambient spillovers “*in the air*”.

Related Literature. We contribute to the longstanding literature on the causes of the dynamic urban wage premium. This builds on the seminal work of [Glaeser and Maré \(2001\)](#), who use survey data to find that a substantial part of the return to working in big cities accumulates over time and is retained when workers move across cities. More recently, [De la Roca and Puga \(2017\)](#) develop a framework to decompose this premium into worker sorting, static agglomeration, and dynamic learning components, using Spanish administrative data to show that workers in bigger cities do not have higher initial unobserved ability. Instead, they accumulate more valuable experience that persists over time and translates to higher wages even after relocating away from the larger cities. [Baum-Snow and Pavan \(2012\)](#) argue that faster wage growth in big cities is driven by greater returns to experience on the job. More recently, [Eckert, Hejlesen, and Walsh \(2022\)](#) use the random allocation of refugees in Denmark to offer causal evidence that the returns to work are higher in bigger cities. [Card, Rothstein, and Yi \(2025\)](#) and [Butts, Jaworski, and Kitchens \(2023\)](#) likewise provide quasi-experimental evidence for the urban wage premium.

Our contribution to this literature is to show that a large share of the dynamic urban wage

premium that has been documented consistently across countries appears to be mediated by differences in firms, coworkers, and job mobility across cities. We build upon and extend the framework of [De la Roca and Puga \(2017\)](#), in a setting where we directly observe the firm at which a worker is employed and the universe of their coworkers. This allows us to account more fully for the relevant inputs to *learning* in addition to production. Our preferred specification nests their dynamic wage equation when expressed in first differences, while allowing us to take account of firm and coworker sorting across space. Our results are not necessarily in tension with prior work that argues for a causal effect of cities on wages. We provide new evidence that the mechanisms behind this effect do not appear to be ambient spillovers but rather the concrete and measurable effects of firm and coworker sorting, along with job mobility.

A large related literature attempts to understand the mechanism behind this dynamic urban wage premium. [Davis and Dingel \(2019\)](#) introduce a dynamic multi-city model where faster growth in bigger cities is a result of costly idea exchange. [Martellini \(2022\)](#), [Crews \(2023\)](#) and [Lhuillier \(2024\)](#) consider models of human capital growth in cities where spillovers depend on the skill distribution of the entire population. [Lindenlaub, Oh, and Peters \(2024\)](#) show that a steeper firm ladder in more productive cities can provide faster wage growth even without any human capital accumulation. Our goal in this paper is to provide suggestive evidence to help guide this theoretical literature, and to provide useful empirical targets for follow-on structural modeling.

Another set of papers are intermediate between the last two literatures, in that they attempt to provide suggestive reduced-form evidence on what mechanisms might be driving higher wages and faster wage growth in cities. [Carry, Kleinman, and Nimier-David \(2025\)](#) use evidence from firm relocations to show that firms and coworkers seem to mediate a large share of the urban wage premium, leaving little room for the effect of the place itself on wages. This builds on the classic analysis of [Combes, Duranton, and Gobillon \(2008\)](#), who show that sorting of workers by itself can explain a large share of regional income differences. [Hong \(2024\)](#), the paper most closely connected to our own, also finds evidence that faster wage growth in cities appears to be mediated by firms and coworkers. We contribute by providing new estimates from French administrative data, studying a richer set of coworker interactions, and providing new evidence on the role of the job ladder on what remains of the dynamic urban growth premium.

Our paper also contributes to the broader literature on human capital accumulation on the job. A burgeoning literature has studied human capital spillovers on the job and found evidence that workers learn from more-skilled coworkers. [Jarosch, Oberfield, and Rossi-](#)

Hansberg (2021), Herkenhoff, Lise, Menzio, and Phillips (2024), and Adenbaum, Babalievsky, and Jungerman (2024) study the sources of human capital growth using structural models and find a major role for learning from higher-skilled coworkers. Nix (2020) finds that having more-educated coworkers causally increases wage growth. Gregory (2023) finds a role for the firms themselves in driving human capital accumulation. We contribute by showing that the mechanisms studied in these papers appear to account for a large share of the dynamic urban wage premium.

1 Data

We use administrative data from France made available to researchers by the French National Statistical Institute, *Institut national de la statistique et des études économiques* (INSEE). These data are widely used in labor and urban economics and allow us to match workers to their employers over time. Specifically, they are compiled from the *Déclarations annuelles de données sociales* (DADS), mandatory tax forms firms in France must submit each year detailing information on their employees (such as wages, hours worked, occupation, etc.). Two main datasets are available. First, for the near-universe of workers in France, we have access to a *short panel* that contains information on all firm-worker-year tuples but scrambles worker identifiers every two years. Second, we use *long panel* which consists of, for a 1-in-12 sample of workers born in October, complete employment histories starting from the worker’s first employment spell.

Sample construction. We apply the same sample restrictions to both the short and the long panels, which we balance. We restrict the sample to private-sector employees aged 18-65 working in metropolitan France.¹ When we use the short panel, we use the 2014-2015 cross-sections and when use the long panel, we use 1997 through 2019. We drop workers with missing information on key variables such wage, gender, and hours, as well as those with non-valid identifiers which we require for longitudinal analysis. We also drop workers from the sample earning less than 100€ per year, but we do not impose any upper limit on wages.

Investigating the role of coworker effects in explaining the urban wage growth premium means we also have to (1) define the boundaries of a city and (2) define the boundaries of a firm or a team, i.e. the set of coworkers.

¹ This excludes workers employed in overseas territories like Martinique and Guadeloupe.

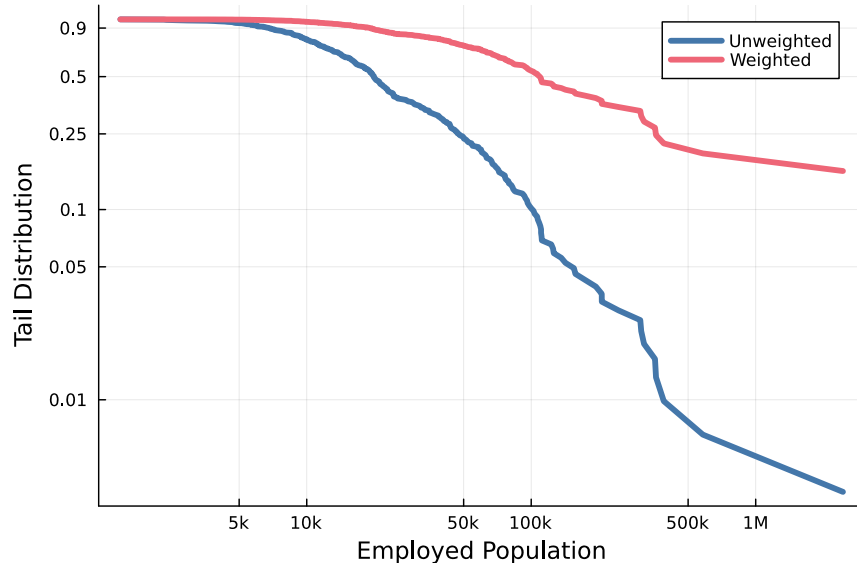


Figure 1: Pareto Tails of the Commuting Zone Size Distribution

Note: This figure plots the complementary cumulative distribution function (tail distribution), defined as the share of firms (establishment $\times$ 1-digit occupation) with more than x employees, for both the unweighted and population-weighted firm size distributions. Computed in the 2015 DADS-Postes.

Commuting zones. Our notion of a city is a commuting zone (CZ), which is a set of municipalities that are economically connected. We use the 2010 version of the CZs, which are defined by INSEE and are based on the 1999 census. The CZs are constructed such that they capture the commuting patterns of workers, with the goal of identifying local labor markets. In Figure 1, we plot the size distribution of CZs in France, measured by employed population. As expected, the long right tail is capturing the dominant role of Paris as the largest labor market in France.

Defining a team. Following the approach in [Adenbaum, Babalievsky, and Jungerman \(2024\)](#), we identify a team as the intersection of an establishment and a 1-digit occupation code. This definition is intended to capture the set of coworkers that a worker is likely to encounter in their day-to-day work environment.

The choice of team boundaries involves an inherent tradeoff. If defined too narrowly, it will exclude interactions between workers who actually collaborate and share knowledge. If defined too broadly, it will incorporate workers who never meaningfully interact. Our definition takes a conservative stance: we treat all workers at a given establishment within the same 1-digit occupation as potential sources of interactions. We do not further partition teams by additional characteristics like industry classification, sector, or echelon (e.g. entry-level versus manager), so that we can take as broad a stance as possible about the potential

learning interactions between workers who may be interacting within an establishment.

Table A.1 presents the 1-digit occupation categories we use, along with selected examples of the more detailed 2-digit and 4-digit codes that fall under *Executives and High-Level Professionals*. If we chose to use finer measures of occupations for our classification of teams, this would imply that workers in closely related roles do not interact. For instance, using either 2 or 4-digit occupation codes would require us to implicitly assume that dentists do not interact in meaningful ways with dental residents, or that lawyers do not interact with other legal professions. Such granular categorizations would artificially exclude what are likely productive learning relationships within the workplace.

Table 1 presents the transition rates across different organizational groupings in our data. Over the 2014 - 2015 period, we observe that 74.11% of workers continue with the same team as we have defined it. Under our definition, there are three possible ways a worker can be classified as changing teams: relocating establishments, switching 1-digit occupations, or doing both at the same time. Transitions across establishments account for the bulk of the mobility we document, with approximately 20% of workers moving between establishments year-over-year (not necessarily within the same firm).

The firm-level self flow rate is much higher than that for establishments, much as we would expect (the firm, after all, is a superset of the establishment). The gap reflects workers who remain with the same employer but switch establishments *within the same firm*. Because these sorts of transitions fundamentally change the composition of colleagues with whom a worker interacts on a day-to-day basis, we classify these individuals as having changed teams even though they remain at the same employer.²

In Table 2, we provide summary statistics for our team definition, both unweighted and employment-weighted, as well as a comparison to alternative definitions. The unweighted average team size in our sample is 4.66 with a 5% of teams being larger than 14 workers. However, the employment weighted mean is 152 with a long right tail. For instance, the 75th percentile is 83 and the 95th percentile is 535. Finally, we compute the mean number of 1-digit occupations per establishment, which is 3, suggesting that establishments on average have a very diverse set of workers.

² We abstract away from the question of whether or not workers are able to interact with their colleagues remotely, since our data window concludes before the onset of the COVID-19 pandemic, and the corresponding widespread adoption of remote working arrangements.

Table 1: Self-Flow Rates

	Rate (%)
OCC1	89.92
Firm	83.64
Establishment	79.16
Establishment $\times$ OCC1	74.11

Note: This table displays the share of workers who remain in the same organizational unit between 2014 and 2015, computed using the DADS-Postes data.

Table 2: Team Definition Summary Statistics

	Mean	SD	p25	Median	p75	p95
Unweighted						
Firm Size	8.30	202.06	1.00	1.00	4.00	20.00
Establishment Size	7.15	43.96	1.00	1.00	4.00	24.00
Establishment $\times$ OCC1 Size	4.66	26.18	1.00	1.00	3.00	14.00
Employment-weighted						
Firm Size	4,925.21	17,113.05	12.00	96.00	930.00	34,005.00
Establishment Size	277.45	1,088.03	8.00	37.00	162.00	986.00
Establishment $\times$ OCC1 Size	151.72	646.95	4.00	17.00	83.00	535.00
OCC1 per Establishment	3.06	1.26	2.00	3.00	4.00	5.00

Note: Summary statistics of team size under our preferred definition (establishment interacted with 1-digit occupation) and alternative organizational boundaries, presented both with and without employment weights. Source: 2015 DADS-Postes.

2 Decomposing the Urban Wage Premium

We now turn to the main analysis. In a first stage, we regress wage growth on commuting zone fixed effects for workers who are employed in both periods of our short panel and stay in the same commuting zone. We estimate

$$w_{i,t} = \nu w_{i,t-1} + \psi_{c(i,t)} + \epsilon_{i,t} \quad (1)$$

where $w_{i,t}$ is the hourly wage of worker i in commuting zone $c(i,t)$ at time t ³, and $\psi_{c(i,t)}$ is a commuting zone fixed effect. The coefficient ν captures the persistence of wages⁴, and the fixed effect $\psi_{c(i,t)}$ captures the static productivity differences across commuting zones. Moreover, we show in Appendix B.1 that this specification formally nests the dynamic specification of De la Roca and Puga (2017) when we restrict attention to workers who have not changed jobs, which will be our preferred specification, as discussed in Section 2.2.⁵ Our main specifications will use hourly wages in levels, however, all of our results are robust to using logged hourly wages as the dependent variable. We report these results in Appendix B.2.

In the second stage, we project down the estimated fixed effects $\hat{\psi}_{c(i,t)}$ from Equation (1) onto the log population $p_{c(i,t)}$ of commuting zone $c(i,t)$:

$$\hat{\psi}_{c(i,t)} = \alpha_0 + \alpha p_{c(i,t)} + \epsilon_{i,t} \quad (2)$$

By projecting the estimated fixed effects onto population in the second stage, we can quantify how much of the cross-commuting zone wage growth variation is systematically related to commuting zone size versus idiosyncratic commuting zone-specific factors. The coefficient α in the second stage tells us the semi-elasticity of hourly wages with respect to city population, while the R^2 reveals what fraction of the cross-city wage differences can be explained by

³ Throughout the paper, we use commuting zone to mean the commuting zone of work. By construction of the commuting zones, residence commuting zone differs from work commuting zone for only a very small fraction of workers.

⁴ Note that this specification allows us to test whether past wages matter for future wage growth. Subtracting by $w_{i,t-1}$ on both sides and rearranging yields

$$w_{i,t} - w_{i,t-1} = \psi_{c(i,t)} + (\nu - 1)w_{i,t-1} + \epsilon_{i,t}$$

If $\nu = 1$, then wages follow a random walk. If $\nu < 1$, then there is mean reversion. Finally, if $\nu > 1$ then the initial advantages associated with being a high-wage worker compound over time.

⁵ We focus on measuring the urban wage *growth* premium, as opposed to the premium in levels. Several papers (Combes, Duranton, and Gobillon, 2008, Carry, Kleinman, and Nimier-David, 2025) have found that the differences in levels are well accounted for by firm and worker sorting, and the results in Glaeser and Maré (2001) and Eckert, Hejlesen, and Walsh (2022) suggest that the static premium is not as robust as the dynamic premium.

population alone. As we progressively add controls in subsequent specifications, changes in both α and R^2 will reveal which mechanisms drive the urban wage premium.

The results for our baseline specification are reported in Column (1) of Table 3. This specification captures the raw urban wage growth premium that we see in the data, without any further controls, and we will refer back to it for the rest of the analysis as we seek to understand which channels and mechanisms drive the effect. In our second stage, we find a semi-elasticity of wages with respect to city size of 0.24. Taken at face value, this would suggest that doubling the population in a commuting zone is correlated with a 0.17€ increase in hourly wage growth, year over year. This is a relatively large effect, but it is almost identical to the results found in De la Roca and Puga (2017) in Spain over a similar time period.⁶ The R^2 of our second stage regression is relatively high: log population alone explains 39.3% of the raw differences in average wage growth across commuting zones.

Next, we add lagged firm fixed effects and estimate the equation

$$w_{i,t} = \psi_{c(i,t)} + \mu_{j(i,t-1)} + \nu w_{i,t-1} + \epsilon_{i,t} \quad (3)$$

where we define the firm $j(i, t - 1)$ to be everyone with the same 1-digit occupation code and the same employer.⁷ The firm fixed effects capture any invariant characteristics of the firms which may be relevant for wage growth at the worker level, including the industry, the type of work they are engaged in, and other invariant characteristics of their workforce composition. For instance, if a firm in our sample consistently has a better composition of coworkers, then this will be absorbed in the firm effect.⁸ As before, in the second stage we project down the estimated commuting zone fixed effects onto log population. The results are reported in Column (2) of Table 3.

⁶ De la Roca and Puga (2017) report an *elasticity* of earnings with respect to population of 0.0241 after controlling for worker fixed effects (Column 4 of Table 1). Our results are not directly comparable, since we report semi-elasticities, but we can do this as a back of the envelope calculation by using the fact that in 2015 in Spain, average earnings were approximately 20,200€ (Arellano, Bonhomme, De Vera, Hospido, and Wei, 2022) and assuming average hours worked per week were 37. This means that the average hourly wage was 10.49€ per hour. This implies a corresponding semi-elasticity of 0.25, which is almost identical to our estimate in Column (1).

⁷ Notice that our definition of a firm here differs from our definition of a team or set of coworkers described in Section 1. We need to use firm IDs and not establishment IDs here because we are already including commuting zone fixed effects. We cannot separately identify establishment times occupation codes fixed effects from commuting zone fixed effects.

⁸ Unlike in Abowd, Kramarz, and Margolis (1999), who aim to jointly identify firm and worker fixed effects, we do not see the full sample of workers for a long enough panel to allow for credible identification of the worker fixed effects. Instead, by including the lagged wage in our specification, we nest the case where we estimate the model in first differences, and therefore are only concerned with identification of the *firm* effects. These are pinned down by the average growth rate of wages within the firm.

Table 3: Decomposing the Urban Wage Premium

	(1)	(2)	(3)	(4)
	Baseline	+ Lagged Firm FE	Coworker Effects	+ Lagged Firm FE
Panel A: First Stage Regressions				
Lagged Wage (ν)	0.8072 (0.0442)	0.6911 (0.0644)	1.0188 (0.0179)	0.9692 (0.0415)
Higher-Wage Coworkers ($\tilde{\theta}_1^+$)			0.1413 (0.0118)	0.1324 (0.0480)
Lower-Wage Coworkers ($\tilde{\theta}_1^-$)			0.1295 (0.0624)	0.0779 (0.0876)
Nonlinear Effects ($\tilde{\theta}_2^+$)			-0.0000 (0.0000)	-0.0000 (0.0000)
Nonlinear Effects ($\tilde{\theta}_2^-$)			-0.0001 (0.0000)	-0.0001 (0.0000)
Lagged Firm Fixed Effects	No	Yes	No	Yes
Observations	15,344,033	13,749,982	13,505,876	13,505,749
R^2	0.1092	0.1318	0.1106	0.1370
Panel B: Second Stage Regressions				
Log Population (α)	0.23614 (0.0278)	0.09086 (0.0210)	0.06361 (0.0131)	0.04877 (0.0141)
Observations	304	304	304	304
R^2	0.3938	0.1171	0.0921	0.0537
$\hat{\alpha}$ change relative to (1)	—	-61.5%	-73.1%	-79.3%
R^2 change relative to (1)	—	-70.3%	-76.6%	-86.4%

Note: Panel A reports first-stage regressions where the dependent variable is hourly wages. Column (1) shows the baseline specification from Equation (1) with commuting zone fixed effects and lagged wages. Column (2) adds firm fixed effects following Equation (3). Column (3) includes the full coworker effects specification from Equation (4). Column (4) combines coworker effects with firm fixed effects following Equation (5). Panel B reports the projection of estimated fixed effects from the first stage onto logged commuting zone population following Equation (2). Standard errors are clustered at the commuting zone level. Sample consists of all workers in the short panel with residualized wages for age and gender.

Having added firm fixed effects, it is striking to examine our measure of the urban wage growth premium (α). We see that the semi-elasticity of hourly wages with respect to commuting zone population falls from 0.24 to 0.09, a drop of 61.5%. By adding firm fixed effects, we account for the fact that the composition of firms varies between big cities and small cities. In Table 4 we regress the employment-weighted average firm fixed effect by commuting zone on log population. We see that high wage growth firms tend to be located in larger cities. This underlying pattern of firm sorting confounds the raw estimates of the urban wage growth premium, and causes it to overstate the importance of population size per se on the growth rate of wages. We see as well that after controlling for firm effects, population becomes dramatically less informative in explaining average changes in wages across commuting zones, only accounting for 11.7% of the variation (compared to 39.3% in our baseline case).

Table 4: Projecting Firm Fixed Effects on Commuting Zone Size

	(1)
Dependent variable	Firm Fixed Effect
Log Population	0.0496 (0.0121)
Observations	304
R^2	0.0530

Note: This table reports regressions of firm fixed effects on commuting zone size.

While firm fixed effects can capture time-invariant firm characteristics, recent evidence suggests that firm-specific features may not be the primary driver of wage dynamics. [Adenbaum, Babalievsky, and Jungerman \(2024\)](#) decompose the sources of human capital growth on the job, and find that peer effects account for more than 50% of the variation in human capital growth for workers in France. Of particular importance is the effect of more skilled coworkers, which contributes 52.5% of the variation in human capital growth, even while accounting for workplace and individual effects. This suggests that who you work with matters more than where you work. Moreover, unlike firm fixed effects, which capture invariant features that are common across all their establishments, the composition of one’s coworkers is quintessentially local. If the composition of coworkers varies substantially across commuting zones, due perhaps to the endogenous mobility choices of workers, small differences in the growth rates

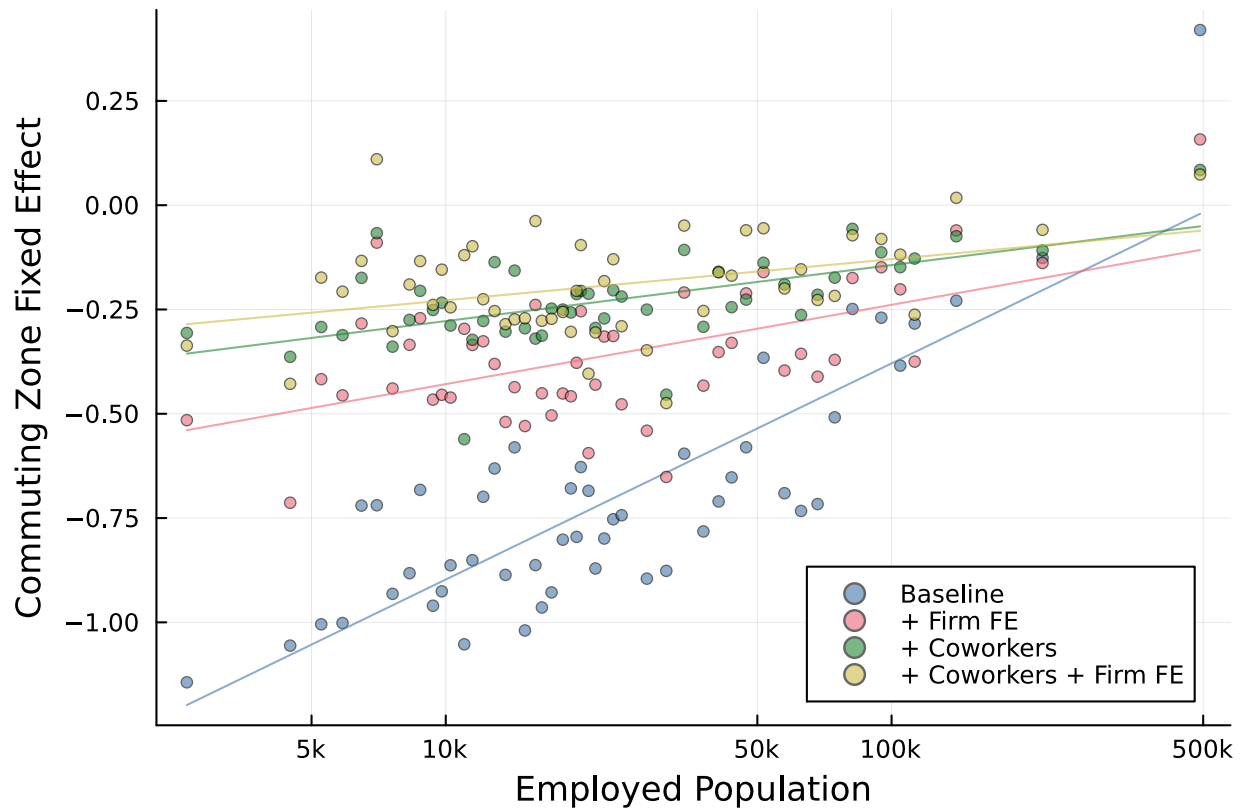


Figure 3: City Fixed Effects vs Population

Note: This figure plots the average commuting zone fixed effects for each 2% quantile bin of commuting zone log population. “Baseline” plots the CZ fixed effects from Equation (1). “Firm FE” plots the CZ fixed effects from Equation (3). “Coworkers” plots the CZ fixed effects from Equation (4). “Coworkers + Firm FE” plots the CZ fixed effects from Equation (5).

of skills across cities, or firm sorting behavior, then we should expect that the opportunities to learn from coworkers will vary with city size.

To test this hypothesis, we augment our baseline specification with controls for peer effects, as in [Adenbaum, Babalievsky, and Jungerman \(2024\)](#) and [Jarosch, Oberfield, and Rossi-Hansberg \(2021\)](#). For each worker i in our sample, we identify the set $\mathbb{W}_{i,t}^+$ of i 's coworkers who are paid more than i , and the set $\mathbb{W}_{i,t}^-$ of coworkers who are paid less than i . We compute over each of these sets the average deviation in wages from worker i 's wage, and the average squared deviation (which captures non-linearity in the learning function).⁹ Consistent with our approach here, we identify the relevant set of coworkers in the previous period $t - 1$, since the wage growth we see between $t - 1$ and t should reflect human capital accumulation in the previous period, with the coworkers at time $t - 1$ rather the present day coworkers. Our choice to separately control for the effects of higher and lower paid workers reflects the findings in [Herkenhoff, Lise, Menzio, and Phillips \(2024\)](#) that the human capital production function is nonlinear, with peer effects that depend on a coworker's relative position to the worker in question. We then estimate the equation

$$\begin{aligned}
w_{i,t} = & \psi_{c(i,t)} + \nu w_{i,t-1} + \underbrace{\tilde{\theta}_1^+ \sum_{j \in \mathbb{W}_{i,t}^+} \frac{w_{j,t-1} - w_{i,t-1}}{n_{k(i)} - 1}}_{\text{Higher-Wage Coworkers}} + \underbrace{\tilde{\theta}_1^- \sum_{j \in \mathbb{W}_{i,t}^-} \frac{w_{j,t-1} - w_{i,t-1}}{n_{k(i)} - 1}}_{\text{Lower-Wage Coworkers}} \\
& + \underbrace{\tilde{\theta}_2^+ \sum_{j \in \mathbb{W}_{i,t}^+} \frac{(w_{j,t-1} - w_{i,t-1})^2}{n_{k(i)} - 1} + \tilde{\theta}_2^- \sum_{j \in \mathbb{W}_{i,t}^-} \frac{(w_{j,t-1} - w_{i,t-1})^2}{n_{k(i)} - 1}}_{\text{Nonlinear Effects}} + \epsilon_{i,t}
\end{aligned} \tag{4}$$

Note that in this regression equation, we do not yet include firm fixed effects. We see that our measure of the urban wage growth premium declines even further in this specification, with the semi-elasticity of hourly wages with respect to commuting zone population falling from 0.24 in our baseline case to 0.06 with coworker effects, a drop of 73.1%. This is of a similar magnitude to the decline that we see in Equation (3), which we interpret as an indication that our coworker controls are capturing much of the same variation as the firm fixed effects. The quality of a firm's workforce (and the corresponding potential for learning from one's coworkers) is common across all the workers at the firm. However, our coworker controls allow us to capture the fact that individual workers within the firm may have different potential for future human capital growth depending on their position within the

⁹ It is not obvious a priori how to compute these coworker controls in a computationally feasible way, since the naive algorithm's cost is quadratic in the size of the dataset. In [Appendix A.2](#) we show how to compute these coworker controls in linear time by carefully rewriting the problem using running sums.

within-firm human capital distribution. For example, the most skilled worker in a firm has no one else to learn from, and our coworker controls can capture this.

In principle, we would like to account for both these sorts of coworker controls *and* for invariant firm effects. To do so, we repeat the coworker specification from Equation (4) adding in lagged firm FE.

$$\begin{aligned}
w_{i,t} = & \psi_{c(i,t)} + \nu w_{i,t-1} + \mu_{j(i,t-1)} + \underbrace{\tilde{\theta}_1^+ \sum_{j \in \mathbb{W}_{i,t}^+} \frac{w_{j,t-1} - w_{i,t-1}}{n_{k(i)} - 1}}_{\text{Higher-Wage Coworkers}} + \underbrace{\tilde{\theta}_1^- \sum_{j \in \mathbb{W}_{i,t}^-} \frac{w_{j,t-1} - w_{i,t-1}}{n_{k(i)} - 1}}_{\text{Lower-Wage Coworkers}} \\
& + \underbrace{\tilde{\theta}_2^+ \sum_{j \in \mathbb{W}_{i,t}^+} \frac{(w_{j,t-1} - w_{i,t-1})^2}{n_{k(i)} - 1} + \tilde{\theta}_2^- \sum_{j \in \mathbb{W}_{i,t}^-} \frac{(w_{j,t-1} - w_{i,t-1})^2}{n_{k(i)} - 1}}_{\text{Nonlinear Effects}} + \epsilon_{i,t}
\end{aligned} \tag{5}$$

It is worth taking a moment to discuss the question of separate identification of the effects here. In principle, we can separately identify coworker and firm effects off of the average growth rate of wages by firm, and the gradient of wage growth *within the firm* against a worker's position within the wage distribution of the firm. That is, if workers who are lower in the wage distribution consistently grow faster than the firm average, then this will be attributed to $\tilde{\theta}_1^+$. In practice, we expect that these effects are likely to be highly co-linear with the firm effects, and difficult to disentangle statistically. However, for our purposes this does not pose a substantial problem: we are not interested in assigning any significant interpretation to the coefficients of our coworker controls. Rather, our object of interest is $\hat{\alpha}$: the correlation between the commuting zone fixed effects $\psi_{c(i,t)}$ and the population of the commuting zone $p_{c(i,t)}$, after partialling out the effects of our other control variables.

This specification also highlights why we do not estimate the full dynamic and static wage specification from [De la Roca and Puga \(2017\)](#) directly, and instead use an estimating equation that coincides with theirs in first differences. Their approach relies on the use of city size bins in order to make the estimation procedure tractable. There is no natural generalization of this to firm fixed effects, or to the coworker learning effects that we are interested in. In particular, it is difficult to imagine how one would capture the *history* of coworker effects in a way that maps neatly onto their experience terms. For this reason, we restrict attention to the specification in first differences, where the generalization is clear.

We can see the results from this specification in column (4) of Table 3. We find that with *both* coworker controls and lagged firm fixed effects, the measured urban wage growth premium is somewhat smaller than with coworker effects alone: the semi-elasticity drops to 0.048,

a 79.3% fall from our baseline. Another way to report these results is to plot the average commuting zone fixed effects for various quantile bins of log population, for each specification. We do this in Figure 3, and we see that the slope of the line of best fit decreases dramatically when we add either firm fixed effects, or coworker controls. The commuting zone fixed effects look fairly similar controlling for either firm fixed effects or coworker fixed effects, suggesting that these two controls are capturing very similar variation in the underlying quality of the learning environments across firms. It is telling that when we compare the fixed effects controlling for coworkers (green) versus coworkers and firm fixed effects (gold), the results are almost indistinguishable to the eye. This reflects the fact that our measures of the urban wage growth premium (α) are statistically indistinguishable between columns (3) and (4) in Table 3.

Robustness. We relegate several robustness checks to the appendix. First, in Appendix B.2 we repeat the analysis for logged hourly wages. The results are reported in Table B.1, respectively. These results show extremely similar qualitative patterns to what we find in our main specification. In fact, in log wages, the results are more striking, with a 98.0% decline relative to our baseline specification when we restrict to job stayers and add coworker controls and firm fixed effects. Second, in Appendix B.3, we repeat the analysis using wage growth as the dependent variable in the first stage. For example, Equation (1) becomes

$$\frac{w_{i,t} - w_{i,t-1}}{w_{i,t} + w_{i,t-1}} \cdot 2 = \psi_{c(i,t)} + \epsilon_{i,t} \quad (1')$$

The results are reported in Table B.4. In these specifications, we find that after accounting for coworker controls, or coworkers controls and firm fixed effects, our measures of the urban wage growth premium are indistinguishable from zero. When we restrict attention only to job stayers, after controlling for coworker controls and firm fixed effects, the sign of the urban wage growth coefficient (α) even becomes negative, although it is not statistically distinguishable from zero.

Finally, in Appendix B.4, we repeat the analysis using alternative measures of the within firm wage distribution to explore how much of our coworker controls can be explained by standard moments. In particular, we consider the mean, variance, skewness, kurtosis, and other percentiles of the within firm wage distribution (the 1st, 10th, 90th, and 99th percentiles). Consistent with our previous specifications, where we allow the effect for a given worker to depend on their relative position within the wage distribution, we introduce all of these controls for the distribution of coworkers and allow for interaction terms with the worker's wage. The results are reported in Table B.6. Even with all of these controls for the distribution of

workers within the firm, our measure of the urban wage growth premium (α) only falls by 42.5% relative to our baseline, compared to 73.1% with our preferred coworker controls. A sensible choice of the specification for coworker controls, motivated by the underlying theory of human capital accumulation within the firm, is key to explaining the variation that we see in the data.

Discussion. These results suggest that insofar as we observe an urban wage growth premium among workers in France, around 80% of the underlying mechanism is mediated through differences in the composition of both firms and coworkers across urban areas. This underlying mechanism in principle is a combination of several things: First, workers could experience wage growth because their skills have increased through learning. Second, workers' wages could increase because they have switched jobs, and this has allowed them to move up the wage ladder. Third, workers wages could have increased in their current job for various non-learning reasons (e.g, if an outside offer gives them leverage to bid up their current wages).

Our learning controls and firm effects speak directly to the first mechanism. However, this approach to measuring the urban wage growth premium conflates the effects of learning with the effects of job to job transitions. If workers transition jobs faster in larger cities, then mechanically we would observe this as a correlation between the commuting zone fixed effects and population. In the next section, we investigate whether or not this is the case: that is, do workers in bigger cities change jobs faster?

2.1 Employment Transitions

To investigate the question of whether or not the job transition rates differ across cities, in Equation (6), we run a simple regression of employer-to-employer (EE) transition rates as a function of log city population:

$$EE_{i,t} = \alpha_0 + \alpha p_{c(i,t)} + \epsilon_{i,t} \quad (6)$$

We use a cross-section from the short panel, which records a worker's employer in the current period and in the lagged period. Note that in this case, our notion of the employer is the same as we use for our definition of the set of coworkers: we identify a worker as experiencing a job transition if they change *establishments* or *1-digit occupations* from one year to the next. This means that we treat both internal movements between establishments within a firm, or changes in roles large enough to change the worker's 1-digit occupation code, as *job*

transitions. For our purposes, such changes in the worker’s team within the firm are very similar to external moves, in terms of the expected effect they will have on wages. We report the results in the first column of Table 5, and find a significant and positive relationship between city size and EE transition rates.

Next, in Equation 7 we add a control for lagged wage, i.e, the wage in the first period of the two-period short panel:

$$EE_{i,t} = \alpha_0 + \alpha p_{c(i,t)} + \beta w_{i,t-1} + \epsilon_{i,t} \quad (7)$$

This regression provides evidence on whether workers leave a job of a given quality more or less quickly. This may be driven by better matching in thicker labor markets, or by the higher prevalence of better jobs in bigger cities that can win poaching fights more easily. We report the results in the second column of Table 5 and again find a significant and positive city size effect. As before, in Table B.3 we repeat the analysis for logged hourly wages and find similar results.

Table 5: Job-to-Job Transition Rates and City Size

	(1)	(2)
Dependent variable	EE Transition	EE Transition
Log Population	0.0296 (0.0017)	0.0368 (0.0012)
Lagged Hourly Wage		-0.0018 (0.0005)
Controls	No	Lagged Wage
Fixed Effects	No	No
SE Clustering	CZ	CZ
Observations	15,344,033	15,344,033
R^2	0.0023	0.0062

Note: This table reports regressions of job-to-job (EE) transition rates on city size. Column (1) shows the baseline specification from Equation (6). Column (2) adds lagged hourly wages as a control following Equation (7). Standard errors are clustered at the commuting zone (CZ) level. The sample consists of all workers from one cross-section of the short panel.

These results stand in contrast to [Martellini \(2022\)](#) and [Hong \(2024\)](#), who find little evidence

for faster job transitions in bigger cities. Understanding the source of this discrepancy is an important goal for future work. It is also important to note that, while the results support the idea that the unexplained remainder of the dynamic urban wage premium might be explained by differences in the rate of EE moves, this is not necessarily evidence of increasing returns to search. As [Lindenlaub, Oh, and Peters \(2024\)](#) show, the returns to EE transitions may be higher because of positive firm sorting towards more productive cities. Hence, while we provide evidence that labor market transitions can help explain what is left of the dynamic urban wage premium after we control for firms and coworkers, more work is still required to understand exactly what drives this faster climb up the job ladder in bigger cities.

2.2 Job stayers

Motivated by our evidence from the previous section that larger cities in France appear to have a higher job-to-job transition rate, we now restrict our attention to the subsample of workers who have not transitioned jobs in the past year. This restriction allows us to separate out the effect of faster transitions from faster skill growth because job stayers, by definition, cannot have moved up on the job ladder within the previous year. Changes in their wages must reflect either learning or bargaining.

We re-run specifications Equations (1) and (3) to (5) on this subsample and report the results in Table 6. Relative to our full sample, we see a similar baseline urban wage growth premium with a measured semi-elasticity of 0.196 relative to 0.236 in the full sample. Adding firm fixed effects, the growth premium falls to 0.068. Controlling for coworker effects it falls to 0.040, and controlling for both it falls to 0.014. This constitutes a remarkable 94.1% drop in the magnitude of the measured urban wage growth premium among job stayers. Moreover, the coefficient is statistically indistinguishable from zero at the 1% level. In Figure 5 we plot the commuting zone fixed effects against log population. As before, we check that these results are robust to a variety of other specifications, reported in Appendix B.2.

So what exactly is “in the air” that makes wages grow faster in bigger cities? Our results provide a surprisingly concrete answer: when we focus on workers who remain in the same job, thereby eliminating the job ladder mechanisms, the urban wage growth premium falls by 94.1% once we account for firms and coworkers. The residual semi-elasticity is not only statistically indistinguishable from zero but also economically negligible. This decomposition suggests that the well-documented urban wage growth premium operates through three specific channels: the quality of firms that locate in cities, the composition of coworkers within those firms, and the speed at which workers can climb the job ladder.

This decomposition helps clarify the mechanisms underlying a phenomenon that has been central to urban economics since [Marshall \(1890\)](#). Rather than diffuse knowledge spillovers that benefit all workers in a city, our findings suggest that faster wage growth is remarkably localized. It depends on the specific firm you work for and the specific people you work with. For job stayers, once we know their firm and coworkers, knowing that they work in Paris versus a small town adds virtually no additional information about their wage growth prospects. This aligns with recent structural evidence on the importance of learning interactions *within the firm*, as in [Adenbaum, Babalievsky, and Jungerman \(2024\)](#), [Herkenhoff, Lise, Menzio, and Phillips \(2024\)](#), and [Jarosch, Oberfield, and Rossi-Hansberg \(2021\)](#). In all of these frameworks, human capital accumulation operates through direct interactions with specific coworkers rather than through ambient effects.

The three channels we identify (firm quality, coworker composition, and job mobility) together substantially account for the observed urban wage growth premium. Our evidence on sorting shows that higher-quality firms concentrate in larger cities, creating a superior learning environment *within the firm*. Meanwhile, our finding of higher EE job transition rates in larger cities suggests that the job ladder operates more quickly there, whether due to thicker labor markets that facilitate better matches or a steeper firm quality distribution as in [Lindenlaub, Oh, and Peters \(2024\)](#). What’s notable is how cleanly these mechanisms decompose: for job stayers, the firm and coworker channels explain virtually everything, while job mobility explains the remainder. Researchers who are interested in modeling the dynamics of wage growth across space need not resort to ambient spillovers operating outside these channels, which are difficult to interpret as structural and microfounded mechanisms.

These findings offer a clearer picture of how urban wage dynamics actually work. By decomposing the urban premium into its constituent parts, we can see that the advantages of cities for wage growth are remarkably concrete and measurable. For workers who remain in the same job, city size itself appears to play no direct role in wage growth once we account for who they work with. While understanding the deeper mechanisms that drive firms and workers to sort into cities, and exactly what generates faster job transitions there, remains an important area for future research, our decomposition provides sharp empirical targets for such work. The mechanisms behind urban wage growth turn out to be more immediate and tangible than the metaphor of knowledge “in the air” might suggest. They operate through the specific people you work with and the specific opportunities available to change jobs. In this fundamental sense, there really is nothing in the air.

Table 6: The Urban Wage Premium Holding Job Mobility Constant

	(1)	(2)	(3)	(4)	(5)
	Baseline (Full Sample)	Baseline (Stayers)	+ Lagged Firm FE	Coworker Effects	+ Lagged Firm FE
Panel A: First Stage Regressions					
Lagged Wage (ν)	0.8072 (0.0442)	0.8610 (0.0636)	0.7461 (0.0973)	0.9856 (0.0209)	0.8728 (0.0573)
Higher-Wage Coworkers ($\tilde{\theta}_1^+$)				0.1389 (0.0156)	0.1634 (0.0542)
Lower-Wage Coworkers ($\tilde{\theta}_1^-$)				-0.0408 (0.0639)	-0.1863 (0.1125)
Nonlinear Effects ($\tilde{\theta}_2^+$)				-0.0000 (0.0000)	-0.0000 (0.0000)
Nonlinear Effects ($\tilde{\theta}_2^-$)				-0.0001 (0.0000)	-0.0001 (0.0000)
Lagged Firm Fixed Effects	No	No	Yes	No	Yes
Observations	15,344,033	9,175,509	8,139,602	8,195,025	8,008,451
R^2	0.1092	0.6849	0.7298	0.7351	0.7710
Panel B: Second Stage Regressions					
Log Population (α)	0.23614 (0.0278)	0.19632 (0.0230)	0.06798 (0.0162)	0.03954 (0.0132)	0.01405 (0.0068)
Observations	304	304	304	304	304
R^2	0.3938	0.2923	0.1253	0.0161	0.0104
$\hat{\alpha}$ change relative to (1)	—	-16.9%	-71.2%	-83.3%	-94.1%
R^2 change relative to (1)	—	-25.8%	-68.2%	-95.9%	-97.4%

Note: Panel A reports first-stage regressions where the dependent variable is hourly wages. Column (1) shows the baseline specification from Equation (1) for the full sample. Columns (2)–(5) restrict to job stayers. Column (2) shows the baseline specification from Equation (1) restricting the sample to job stayers. Column (3) adds firm fixed effects following Equation (3). Column (4) includes the full coworker effects specification from Equation (4). Column (5) combines coworker effects with firm fixed effects following Equation (5). Panel B reports the projection of estimated fixed effects from the first stage onto logged commuting zone population following Equation (2). Standard errors are clustered at the commuting zone level. Wages are residualized for age and gender.

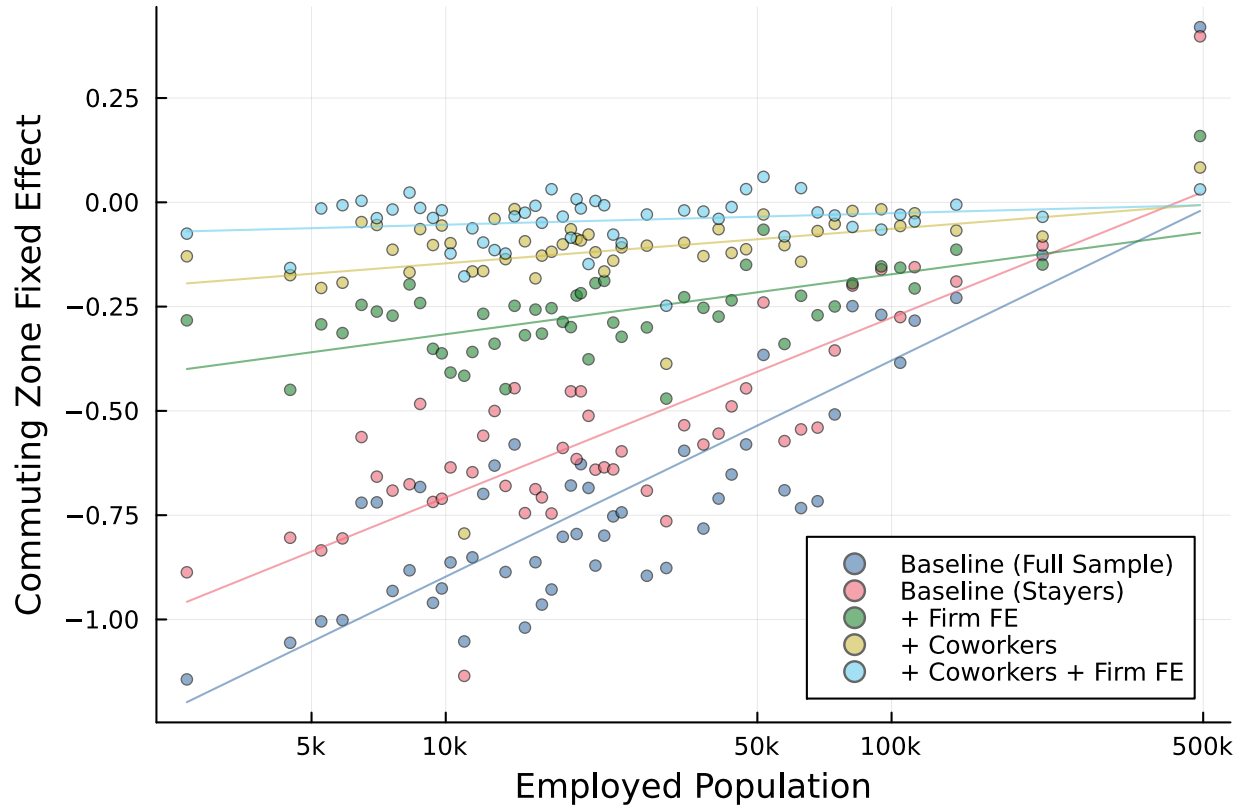


Figure 5: City Fixed Effects vs Population for Job Stayers

Note: This figure plots the average commuting zone fixed effects for each 2% quantile bin of commuting zone log population, in our sample restricted to job stayers. “Baseline (Full Sample)” plots the CZ fixed effects from Equation (1) for the full sample. “Baseline (Stayers)” plots the CZ fixed effects from Equation (1) when we restrict to job stayers. “Firm FE” plots the CZ fixed effects from Equation (3) for job stayers. “Coworkers” plots the CZ fixed effects from Equation (4) for job stayers. “Coworkers + Firm FE” plots the CZ fixed effects from Equation (5) for job stayers.

Conclusion

In this paper, we have used French administrative data to provide new evidence on the causes of the dynamic urban wage premium. We first demonstrated that this dynamic premium is substantially attenuated after adding firm fixed effects, suggesting that part of this premium is driven by the firms that choose to sort themselves into bigger cities. Next, we showed that the premium shrinks even further when adding controls for coworker wages, which suggests a role for learning from the higher human capital workers that sort themselves into cities. Crucially, we showed that moments of the worker wage distribution are not sufficient in themselves—what matters is the mass of coworkers with higher and lower wages than the reference worker. Combining firm and coworker controls greatly attenuated the dynamic urban wage premium in our full sample, and eliminated it altogether in our subsample of job stayers. Motivated by this discrepancy between job stayers and the full sample, we showed that employer-to-employer transition rates are higher in bigger French cities. This suggests that labor market factors (whether they are increasing returns to search or a steeper firm quality ladder) help explain the remaining part of the dynamic urban wage premium that survived our worker and firm controls. Taken at face value, our results leave little room for the canonical explanation for the dynamic urban wage premium, namely human capital spillovers “*in the air*”.

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Appendix

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A Data Appendix

A.1 Variable List

- Hourly Wages: We use hourly wages to control for variation in hours worked. Wages are `S_BRUT` in the short panel and `SB` in the long panel. These are gross annual wages which include overtime and bonuses. We divide these by `NBHEUR`, the number of hours worked in the year, to get hourly wages. Finally, we use the CPI provided by INSEE to deflate all wages to 2015 euros.
- Establishment: We use the INSEE provided establishment codes `SIRET` to define the boundaries of a team.
- Occupation: We also use 1-digit occupation codes, the first digit of `PCS4`, to define a team.

Table A.1: Occupation Coding in DADS Data

1	Farmers
2	Craftsmen, Tradespeople, and Business Owners
3	Executives and High-Level Professionals
31	<i>Independent Professionals</i>
311c	Dentists
311d	Psychologists and Therapists
311e	Veterinarians
3121	Lawyers
33	<i>Public Sector Executives</i>
34	<i>Professors, Scientific Professionals</i>
342b	Research Professors
344a	Hospital Doctors Without an Independent Practice
344c	Residents in Medicine, Dentistry and Pharmacy
344d	Salaried Pharmacists
35	<i>Careers in Media, Arts, and Entertainment</i>
352a	Journalists
352b	Literary Authors, Screenwriters
353a	Newspaper Editors, Media Executives, Publishing Directors
37	<i>Corporate Administrative and Commercial Managers</i>
372e	Legal Professionals
375a	Advertising Executives
376a	Financial Market Executives
38	<i>Engineers and Technical Managers</i>
382b	Salaried Architects
384b	Mechanical Manufacturing Engineers and Metalworking Managers
387c	Production Process Engineers and Managers
387d	Quality Control Engineers and Managers
4	Intermediate Professions
5	Clerical Workers
6	Manual Laborers
9	Non-Coded

Note: This table illustrates occupation coding in the DADS. There are 6 distinct 1-digit occupation codes. To illustrate the granularity of 2-digit occupation coding, we also report a selection of the 2-digit and 4-digit codes for a representative example. The full list of occupation codes can be found on INSEE's website at <https://www.insee.fr/fr/information/2497952>.

A.2 Quadratic Expansion

In this appendix, we derive computationally efficient formulas for the positive and negative components of the coworker effects in both the linear and quadratic cases, which we use to estimate Equations (4) and (5). These formulas can be computed in linear time after sorting the data, and do not require any look-aheads or look-backs through the data. We begin with the linear case. Suppose we have a sequence $\{x_i\}_{i=1}^N$ and we want to compute, for each $j \in \{1, 2, \dots, N\}$

$$y_j^+ := \frac{1}{N} \sum_{i=1}^N \max\{x_i - x_j, 0\}$$

Without loss of generality, we will assume throughout that the sequence $\{x_i\}$ is sorted in ascending order, i.e. $x_1 \leq x_2 \leq x_3 \leq \dots \leq x_N$. Moreover, we use Z_j to denote the running sum

$$Z_j := \sum_{i=1}^j x_i$$

which can conveniently be computed in a single pass through the data (i.e in linear time) since $Z_{j+1} = Z_j + x_{j+1}$. Using this, we can rewrite y_j^+ as

$$\begin{aligned} y_j^+ &= \frac{1}{N} \sum_{i=j+1}^N (x_i - x_j) && (x_i \geq x_j \text{ for } i \geq j) \\ &= \frac{1}{N} \left[\sum_{i=j+1}^N x_i - (N-j)x_j \right] \\ &= \frac{1}{N} [Z_N - Z_j - (N-j)x_j] \end{aligned}$$

We can also do the same for the negative component

$$\begin{aligned}
y_j^- &:= \frac{1}{N} \sum_{i=1}^N \min\{x_i - x_j, 0\} \\
&= \frac{1}{N} \sum_{i=1}^{j-1} (x_i - x_j) && (x_i \geq x_j \text{ for } i \geq j) \\
&= \frac{1}{N} [Z_{j-1} - (j-1)x_j] \\
&= \frac{1}{N} [Z_j - x_j - (j-1)x_j] \\
&= \frac{1}{N} (Z_j - jx_j)
\end{aligned}$$

We now turn to the quadratic terms and similarly define

$$Z_{2,j} := \sum_{i=1}^j x_i^2$$

to be the running sum of the x_i^2 terms up to j . The positive component in the quadratic case can then be rewritten as

$$\begin{aligned}
y_{2,j}^+ &= \frac{1}{N} \sum_{i=1}^N \max\{x_i - x_j, 0\}^2 \\
&= \frac{1}{N} \sum_{i=j+1}^N (x_i - x_j)^2 \\
&= \frac{1}{N} \sum_{i=j+1}^N (x_i^2 - 2x_i x_j + x_j^2) \\
&= \frac{1}{N} \left[\sum_{i=j+1}^N x_i^2 - 2x_j \sum_{i=j+1}^N x_i + (N-j)x_j^2 \right] \\
&= \frac{1}{N} [(Z_{2,N} - Z_{2,j}) - 2x_j(Z_N - Z_j) + (N-j)x_j^2]
\end{aligned}$$

Similarly, for the negative component, we have

$$\begin{aligned}
y_{2,j}^- &= \frac{1}{N} \sum_{i=1}^N \min\{x_i - x_j, 0\}^2 \\
&= \frac{1}{N} \sum_{i=1}^{j-1} (x_i - x_j)^2 \\
&= \frac{1}{N} \sum_{i=1}^{j-1} (x_i^2 - 2x_i x_j + x_j^2) \\
&= \frac{1}{N} \left[\sum_{i=1}^{j-1} x_i^2 - 2x_j \sum_{i=1}^{j-1} x_i + (j-1)x_j^2 \right] \\
&= \frac{1}{N} [Z_{2,j-1} - 2x_j Z_{j-1} + (j-1)x_j^2] \\
&= \frac{1}{N} [(Z_{2,j} - x_j^2) - 2x_j(Z_j - x_j) + (j-1)x_j^2] \\
&= \frac{1}{N} [Z_{2,j} - 2x_j(Z_j - x_j) + (j-2)x_j^2]
\end{aligned}$$

B Results Appendix

B.1 Relationship to De la Roca and Puga (2017)

In this section, we show that our specification in Equation (1) restricted to job stayers nests the dynamic specification from De la Roca and Puga (2017) in first differences. To begin, let us start with their dynamic specification (Equation 1, Section 3, p. 113). They posit a model of wages where

$$w_{it} = \sigma_{c(i,t)} + \mu_i + \sum_{j=1}^C \delta_{j,c(i,t)} e_{ijt} + \mathbf{x}'_{it} \beta + \varepsilon_{it} \quad (\text{B.1})$$

If we take first differences:

$$\Delta w_{it} = \sigma_{c(i,t)} - \sigma_{c(i,t-1)} + \left[\sum_{j=1}^C \delta_{j,c(i,t)} e_{ijt} - \sum_{j=1}^C \delta_{j,c(i,t-1)} e_{ij,t-1} \right] + \Delta \mathbf{x}'_{it} \beta + \Delta \varepsilon_{it} \quad (\text{B.2})$$

In our stayers sample, we know that, by construction $c(i, t) = c(i, t-1)$ so $\sigma_{c(i,t)} - \sigma_{c(i,t-1)} = 0$. Moreover, we also know that $\delta_{j,c(i,t)} = \delta_{j,c(i,t-1)}$. So, for our stayers sample we can simplify Equation (B.3) to

$$\Delta w_{it} = \sum_{j=1}^C \delta_{j,c(i,t)} \Delta e_{ijt} + \Delta \mathbf{x}'_{it} \beta + \Delta \varepsilon_{it} \quad (\text{B.3})$$

But, the structure of e_{ijt} , which is the cumulative number of years that worker i has spent in city j up until time t means that $\Delta e_{ijt} = e_{ijt} - e_{ij,t-1} = \mathbb{1}\{j = c(i, t)\}$. This means that the terms in the summand drop out for all $j \neq c(i, t)$, and we obtain

$$\Delta w_{it} = \delta_{c(i,t), c(i,t)} + \Delta \mathbf{x}'_{it} \beta + \Delta \varepsilon_{it} \quad (\text{B.4})$$

So, when we estimate:

$$w_{i,t} = \nu w_{i,t-1} + \psi_{c(i,t)} + \epsilon_{i,t} \quad (1)$$

in our sample of job stayers, our firm commuting zone fixed effects $\psi_{c(i,t)}$ sweep out the δ terms, and our inclusion of the lagged wage $w_{i,t-1}$ nests Equation (B.4) in the case where $\nu = 1$.¹⁰ In our stayers regressions, the coefficients that De la Roca and Puga (2017) identify as the dynamic effects that accrue to experience in each city type are captured in the commuting zone fixed effects $\psi_{c(i,t)}$.

¹⁰ Note: many of the worker level controls that De la Roca and Puga (2017) include in $\mathbf{x}_{it}$ are also largely swept out when taking first differences (for instance, the linear components of experience and job tenure). In our specifications we also add fixed effects for observable worker characteristics such as gender and age.

B.2 Robustness to Log Wages

Table B.1: Decomposing the Urban Wage Premium: Log Hourly Wages

	(1)	(2)	(3)	(4)
	Baseline	+ Lagged Firm FE	Coworker Effects	+ Lagged Firm FE
Panel A: First Stage Regressions				
Lagged Wage (ν)	0.8313 (0.0005)	0.5590 (0.0010)	0.9141 (0.0003)	0.7706 (0.0020)
Higher-Wage Coworkers ($\tilde{\theta}_1^+$)			0.2299 (0.0018)	0.0724 (0.0027)
Lower-Wage Coworkers ($\tilde{\theta}_1^-$)			0.0315 (0.0013)	-0.0357 (0.0032)
Nonlinear Effects ($\tilde{\theta}_2^+$)			0.0770 (0.0009)	0.0784 (0.0010)
Nonlinear Effects ($\tilde{\theta}_2^-$)			-0.0064 (0.0008)	-0.0240 (0.0001)
Lagged Firm Fixed Effects	No	Yes	No	Yes
Observations	15,344,033	13,749,982	13,505,876	13,505,749
R^2	0.7531	0.7667	0.7621	0.7940
Panel B: Second Stage Regressions				
Log Population (α)	0.00813 (0.0008)	0.00351 (0.0008)	0.00327 (0.0004)	0.00213 (0.0007)
Observations	304	304	304	304
R^2	0.3756	0.0866	0.1908	0.0457
$\hat{\alpha}$ change relative to (1)	—	-56.8%	-59.9%	-73.8%
R^2 change relative to (1)	—	-76.9%	-49.2%	-87.8%

Note: Panel A reports first-stage regressions where the dependent variable is log hourly wages. Column (1) shows the baseline specification from Equation (1) with commuting zone fixed effects and lagged wages. Column (2) adds firm fixed effects following Equation (3). Column (3) includes the full coworker effects specification from Equation (4). Column (4) combines coworker effects with firm fixed effects following Equation (5). Panel B reports the projection of estimated fixed effects from the first stage onto logged commuting zone population following Equation (2). Standard errors are clustered at the commuting zone level. Sample consists of all workers in the short panel with residualized wages for age and gender.

Table B.2: The Urban Wage Premium Holding Job Mobility Constant: Log Hourly Wages

	(1)	(2)	(3)	(4)	(5)
	Baseline (Full Sample)	Baseline (Stayers)	+ Lagged Firm FE	Coworker Effects	+ Lagged Firm FE
Panel A: First Stage Regressions					
Lagged Wage (ν)	0.8313 (0.0005)	0.9526 (0.0004)	0.8068 (0.0013)	0.9768 (0.0003)	0.9529 (0.0024)
Higher-Wage Coworkers ($\tilde{\theta}_1^+$)				0.1376 (0.0024)	0.1151 (0.0034)
Lower-Wage Coworkers ($\tilde{\theta}_1^-$)				0.0238 (0.0013)	0.0499 (0.0043)
Nonlinear Effects ($\tilde{\theta}_2^+$)				0.0638 (0.0016)	0.0686 (0.0018)
Nonlinear Effects ($\tilde{\theta}_2^-$)				-0.0064 (0.0008)	0.0131 (0.0019)
Lagged Firm Fixed Effects	No	No	Yes	No	Yes
Observations	15,344,033	9,175,509	8,139,602	8,195,025	8,008,451
R^2	0.7531	0.8906	0.9042	0.8888	0.9076
Panel B: Second Stage Regressions					
Log Population (α)	0.00813 (0.0008)	0.00302 (0.0004)	0.00141 (0.0005)	0.00125 (0.0003)	0.00016 (0.0004)
Observations	304	304	304	304	304
R^2	0.3756	0.1757	0.0351	0.0324	0.0006
$\hat{\alpha}$ change relative to (1)	—	-62.9%	-82.7%	-84.6%	-98.0%
R^2 change relative to (1)	—	-53.2%	-90.7%	-91.4%	-99.8%

Note: Panel A reports first-stage regressions where the dependent variable is log hourly wages. Column (1) shows the baseline specification from Equation (1) for the full sample. Columns (2)-(5) restrict to job stayers. Column (2) shows the baseline specification from Equation (1) restricting the sample to job stayers. Column (3) adds firm fixed effects following Equation (3). Column (4) includes the full coworker effects specification from Equation (4). Column (5) combines coworker effects with firm fixed effects following Equation (5). Panel B reports the projection of estimated fixed effects from the first stage onto logged commuting zone population following Equation (2). Standard errors are clustered at the commuting zone level. Wages are residualized for age and gender.

Table B.3: Job-to-Job Transition Rates and City Size: Log Hourly Wages

	(1)	(2)
Dependent variable	EE Transition	EE Transition
Log Population	0.0296 (0.0017)	0.0468 (0.0030)
Lagged Log Wage		-0.1032 (0.0014)
Controls	No	Lagged Wage
Fixed Effects	No	No
SE Clustering	CZ	CZ
Observations	15,344,033	15,344,033
R^2	0.0023	0.0175

Note: This table reports regressions of job-to-job (EE) transition rates on city size. Column (1) shows the baseline specification from Equation (6). Column (2) adds lagged log hourly wages as a control following Equation (7). Standard errors are clustered at the commuting zone (CZ) level. The sample consists of all workers from one cross-section of the short panel.

B.3 Robustness to Wage Growth

Table B.4: Decomposing the Urban Wage Premium: Wage Growth

	(1)	(2)	(3)	(4)
	Baseline	+ Lagged Firm FE	Coworker Effects	+ Lagged Firm FE
Panel A: First Stage Regressions				
Higher-Wage Coworkers ($\tilde{\theta}_1^+$)			0.0076 (0.0001)	0.0155 (0.0003)
Lower-Wage Coworkers ($\tilde{\theta}_1^-$)			0.0025 (0.0001)	0.0004 (0.0001)
Nonlinear Effects ($\tilde{\theta}_2^+$)			-0.0000 (0.0000)	-0.0000 (0.0000)
Nonlinear Effects ($\tilde{\theta}_2^-$)			0.0000 (0.0000)	0.0000 (0.0000)
Lagged Firm Fixed Effects	No	Yes	No	Yes
Observations	15,344,033	13,749,982	13,505,876	13,505,749
R^2	0.0123	0.1462	0.0453	0.1953
Panel B: Second Stage Regressions				
Log Population (α)	0.00128 (0.0003)	0.00135 (0.0005)	0.00054 (0.0003)	0.00084 (0.0005)
Observations	304	304	304	304
R^2	0.0594	0.0327	0.0098	0.0122
$\hat{\alpha}$ change relative to (1)	—	+5.2%	-58.0%	-34.2%
R^2 change relative to (1)	—	-45.0%	-84.8%	-79.5%

Note: Panel A reports first-stage regressions where the dependent variable is hourly wage growth. Column (1) shows the baseline specification from Equation (1') with commuting zone fixed effects. Similarly, columns (2)-(4) report the results from the specifications where we change Equations (3) to (5) to use wage growth as the dependent variable. Panel B reports the projection of estimated fixed effects from the first stage onto logged commuting zone population following Equation (2). Standard errors are clustered at the commuting zone level. Sample consists of all workers in the short panel with residualized wages for age and gender.

Table B.5: Job Stayers with Wage Growth as Dependent Variable

	(1)	(2)	(3)	(4)	(5)
	Baseline (Full Sample)	Baseline (Stayers)	+ Lagged Firm FE	Coworker Effects	+ Lagged Firm FE
Panel A: First Stage Regressions					
Higher-Wage Coworkers ($\tilde{\theta}_1^+$)				0.0390 (0.0008)	0.0083 (0.0002)
Lower-Wage Coworkers ($\tilde{\theta}_1^-$)				0.0010 (0.0000)	-0.0001 (0.0000)
Nonlinear Effects ($\tilde{\theta}_2^+$)				-0.0000 (0.0000)	-0.0000 (0.0000)
Nonlinear Effects ($\tilde{\theta}_2^-$)				0.0000 (0.0000)	-0.0000 (0.0000)
Lagged Firm Fixed Effects	No	No	Yes	No	Yes
Observations	15,344,033	9,175,509	8,139,602	8,195,025	8,008,451
R^2	0.0123	0.0081	0.1886	0.0266	0.2154
Panel B: Second Stage Regressions					
Log Population (α)	0.00128 (0.0003)	0.0007805 (0.000279)	0.00003 (0.0002985)	0.0002909 (0.0003099)	-0.0003749 (0.0003282)
Observations	304	304	304	304	304
R^2	0.0594	0.0194	0.0000	0.0021	0.0043
$\hat{\alpha}$ change relative to (1)	—	-39.0%	-97.7%	-77.3%	-129.3%
R^2 change relative to (1)	—	-67.3%	-100.0%	-96.5%	-92.8%

Note: Panel A reports first-stage regressions where the dependent variable is hourly wage growth. Column (1) shows the baseline specification from Equation (1') with commuting zone fixed effects. Columns (2)-(5) restrict to job stayers. Column (2) shows the baseline specification from Equation (1') restricting the sample to job stayers. Similarly, columns (2)-(4) report the results from the specifications where we change Equations (3) to (5) to use wage growth as the dependent variable. Panel B reports the projection of estimated fixed effects from the first stage onto logged commuting zone population following Equation (2). Standard errors are clustered at the commuting zone level. Wages are residualized for age and gender.

B.4 Other Moments

Table B.6: Adding Moments of the Coworker Wage Distribution

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
	Baseline	+Mean	+Variance	+Skewness	+Kurtosis	+P1	+P10	+P90	+P99
Panel A: First Stage Regressions									
Lagged Wage	0.8072 (0.0442)	1.0260 (0.0913)	0.9649 (0.0920)	0.8916 (0.1068)	0.8968 (0.1126)	0.7947 (0.1081)	0.7140 (0.0858)	0.6802 (0.0773)	0.6900 (0.0757)
Mean Wage		-0.1713 (0.1356)	-0.1007 (0.1397)	-0.0837 (0.1466)	-0.0857 (0.1474)	-0.1688 (0.1539)	-0.3586 (0.1729)	-0.4858 (0.1529)	-0.5065 (0.1541)
Mean Wage $\times$ Lagged Wage		-0.0009 (0.0002)	-0.0012 (0.0003)	-0.0011 (0.0003)	-0.0011 (0.0003)	-0.0011 (0.0003)	-0.0009 (0.0003)	-0.0007 (0.0001)	-0.0007 (0.0001)
Variance			0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	-0.0000 (0.0000)	-0.0000 (0.0000)
Variance $\times$ Lagged Wage			0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)	0.0000 (0.0000)
Skewness				-0.1129 (0.1210)	0.1150 (0.2096)	-0.2209 (0.1748)	-0.5306 (0.1662)	-0.6622 (0.1501)	-0.7848 (0.2140)
Skewness $\times$ Lagged Wage				0.0150 (0.0043)	0.0121 (0.0082)	0.0253 (0.0075)	0.0315 (0.0084)	0.0349 (0.0074)	0.0344 (0.0095)
Kurtosis					-0.0099 (0.0040)	-0.0021 (0.0032)	0.0003 (0.0030)	0.0063 (0.0031)	0.0085 (0.0048)
Kurtosis $\times$ Lagged Wage					0.0001 (0.0001)	-0.0000 (0.0001)	-0.0003 (0.0001)	-0.0003 (0.0001)	-0.0003 (0.0002)
P1						0.3272 (0.0768)	-0.1359 (0.0944)	-0.0535 (0.0859)	-0.0232 (0.1004)
P1 $\times$ Lagged Wage						0.0019 (0.0005)	0.0008 (0.0035)	-0.0008 (0.0031)	-0.0011 (0.0035)
P10							0.9029 (0.2190)	0.6541 (0.1827)	0.6599 (0.1844)
P10 $\times$ Lagged Wage							0.0007 (0.0032)	0.0024 (0.0035)	0.0024 (0.0036)
P90								0.1919 (0.0380)	0.1491 (0.0323)
P90 $\times$ Lagged Wage								-0.0004 (0.0006)	-0.0001 (0.0003)
P99									0.0218 (0.0074)
P99 $\times$ Lagged Wage									-0.0001 (0.0001)
Observations	15,344,033	13,505,876	13,149,735	13,139,708	13,139,708	13,139,708	13,139,708	13,139,708	13,139,708
R^2	0.1092	0.1098	0.1270	0.1280	0.1280	0.1314	0.1360	0.1619	0.1628
Panel B: Second Stage Regressions									
Log Population (α)	0.23614 (0.0278)	0.2413 (0.0301)	0.24697 (0.0296)	0.24220 (0.0277)	0.23448 (0.0277)	0.26442 (0.0304)	0.15896 (0.0236)	0.12458 (0.0127)	0.13569 (0.0121)
$\hat{\alpha}$ change relative to (1)	—	+2.2%	+4.6%	+2.6%	-0.7%	+12.0%	-32.7%	-47.2%	-42.5%
R^2 change relative to (1)	—	-5.2%	-4.0%	-6.2%	-9.4%	-25.3%	-17.6%	-1.8%	+2.0%

Note: Panel A reports first-stage regressions where the dependent variable is hourly wages. All specifications include commuting zone fixed effects, age, and gender controls. Column (1) includes lagged wages. Column (2) adds mean wage and its interaction with lagged wage. Column (3) adds variance and its interaction with lagged wage. Column (4) adds skewness and its interaction. Column (5) adds kurtosis and its interaction. Column (6) adds the 1st percentile (P1) and its interaction. Column (7) adds the 10th percentile (P10) and its interaction. Column (8) adds the 90th percentile (P90) and its interaction. Column (9) adds the 99th percentile (P99) and its interaction. Panel B reports the projection of estimated commuting zone fixed effects from the first stage onto logged commuting zone population. Robust standard errors reported. Sample consists of all workers in the short panel with residualized wages for age and gender.